Empirical Variational Bayes Factorization for Longitudinal Matrix Data

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Abstract

Collecting high-dimensional data over time for multiple individuals is increasingly common in biomedical research, particularly in studies of disease progression or molecular time-course experiments. Often data are missing or the timepoints are irregularly spaced across individuals. There is a lack of flexible methodology to explore latent patterns of change over time and account for missing data. We propose an empirical variational Bayes (EVB) approach to factorize high-dimensional longitudinal data, in which the underlying factors have temporal correlation over time. The EVB approach is computationally efficient and accommodates flexible covariance structures over time with automatic determination of all hyper-parameters (including the underlying rank, i.e., number of factors). Simulations demonstrate that the approach captures the underlying signal more precisely than other strategies, and performs well with respect to missing data imputation under different scenarios. An application to a gene time-course data for multiple sclerosis treatment provides a factorization with interpretable components over time and improved imputation accuracy for missing data.

Keywords: Dimension reduction, Gaussian process, Low-rank approximation, Temporal correlation, Variational Bayes

1. Introduction

Matrix factorization methods have long been critical to analyze and summarize high-dimensional data for diverse tasks such as latent structure inference, data compression, and missing data imputation. Recently, empirical variational Bayes (EVB) has been shown to be a powerful, flexible, and efficient approach to low-rank matrix factorization. Under a variational Bayes

approach the true posterior for a Bayesian model is approximated by a variational distribution with simplified assumptions [1], and under EVB the model hyperparameters are also empirically determined. Previous work describes a simple EVB approach to matrix factorization that results in an efficient closed-form low-rank solution [2]. Empirical Bayes matrix factorization flexibly incorporates different forms of priors, allowing novel combinations of classical models such as sparse matrix factorization and non-negative matrix factorization [3]. In the context of multiple linked matrices, EVB has been used to decompose signal into low-rank approximations for shared and unshared structure [4]. When auxiliary covariates are available, empirical Bayes methods can incorporate this information directly into prior estimation [5].

Advancements in biotechnology have enabled high-throughput data collection (e.g., for "omics" or imaging modalities) from multiple timepoints, motivating longitudinal studies. However, in most cases it is difficult to obtain complete data, often caused by patient unavailability or dropout. This can lead to several issues, such as biased results and a reduction in statistical power, compromising the validity of the study and the reliability of its findings. As a motivating example, consider a study that investigates genomic responses of multiple sclerosis (MS) patients to the genetic therapy recombinant human interferon beta (rIFNβ) [6]. The study involves analyzing a gene time course dataset that measures 76 gene expression levels in 53 subjects, at 7 timepoints over a 2-year followup period. However, 10.5% of the data are missing across all timepoints, limiting the validity of inference.

Several classes of methods have been developed to apply factorization models to longitudinal data settings. One popular area of research are tensor factorization methods [7]. The tensor framework generalizes the two-way matrix to higher-order arrays. Tensor factorization methods enjoy many desirable properties of matrix factorization, such as low rank representation of the data and interpretable additive components. They can conveniently accommodate longitudinally collected high-dimensional matrices by adding time as a third dimension, forming a third-order tensor array. Tensor factorization methods have been developed specifically to accommodate temporal structure, by incorporating functional representations or smoothing along the temporal dimension [8, 9]. Other tensor methods have been developed for imputation in a longitudinal setting using a tensor factorization [10]. However, treating time as an additional dimension within a traditional tensor framework assumes that timepoints are aligned across individuals.

Another class of methods extend matrix factorization models to longitu-

dinal settings. For example, [11] use a factor analysis approach and models the scores using a multivariate linear mixed model. This induces correlation within individual observations, but limits flexibility to model the correlation to the specific form under a linear mixed model. Bayesian methods, on the other hand, can deal with more complex correlation structures by naturally incorporating flexible priors for the factors. For example, [12] uses a factor model for random effects within a mixed modeling framework, where the scores are employed with an infinite mixture prior. However, this work centered on the task of clustering instead of signal estimation or imputation, which is the focus of this paper. Alternatively, [13] propose a longitudinal factor model for disease progression from high-dimensional data, but assume a transition between two latent states over time rather than a more general correlation structure.

We propose an empirical variation Bayes factor model for high-dimensional longitudinal data that simultaneously addresses estimation and imputation, even when timepoints are inconsistently measured. Our contribution has four main components. First, we introduce a general repeated measures model for high-dimensional data using latent factors that flexibly accommodates a range of correlation structures over time. Second, we use an EVB framework for estimation that does not require *any* user-specified tuning parameters, and that achieves automatic rank selection—an important advantage since manual rank selection can be difficult or computationally prohibitive for large datasets. Third, we develop a novel empirical Bayes random process model (EBRPM), which extends prior work to allow efficient computation and unified estimation of both priors and variationally approximated posteriors. This formulation leverages empirical Bayes principles to maximize efficiency while retaining the ability to recover complex signals in high-dimensional settings. Fourth, our framework naturally handles missing data, including both entry-wise and blockwise missingness, by retaining a key Bayesian property: the sharing of information across individuals to estimate a common prior distribution. This feature is especially important for blockwise missingness, where data for an individual may be entirely absent at a timepoint and imputation without borrowing information and temporal correlation is impossible.

2. Methods

2.1. Model Construction

Here we describe our model for aligned measurement times across all subjects; extensions to scenarios with irregularly measured timepoints across subjects are straightforward, and are described in Appendix A.

Assume there are n subjects, each with p variables measured at $t_1, \ldots, t_D$ timepoints. We define the rank R model as

$$\boldsymbol{Y}_t = \boldsymbol{U}\boldsymbol{V}_t^T + \boldsymbol{E}_t,\ t = t_1, \ldots, t_D \tag{1}$$

where $\boldsymbol{Y}_t \in \mathbb{R}^{p\times n}$ are the observed data at each timepoint, $\boldsymbol{U} \in \mathbb{R}^{p\times R}$, $\boldsymbol{V}_t \in \mathbb{R}^{n\times R}$ are factor matrices and $\boldsymbol{E}_t$ is an error matrix at time t. Following standard terminology, we refer to $\boldsymbol{U}$ as the *loading* matrix, $\boldsymbol{V}$ as the *score* matrix, and R as the rank/number of factors. The model can be expanded as the sum R rank-1 components, via the outer products between columns of $\boldsymbol{U}$ and $\boldsymbol{V}_t$,

$$\boldsymbol{Y}_t = \sum_{r=1}^{R} \boldsymbol{u}_r \boldsymbol{v}_{tr}^T + \boldsymbol{E}_t \tag{2}$$

where $\boldsymbol{u}_r$ and $\boldsymbol{v}_{tr}$ denote the r-th column of $\boldsymbol{U}$ and $\boldsymbol{V}_t$, respectively. Following a Bayesian framework, we give independent normal priors to each column of $\boldsymbol{U}$, and a joint normal prior to $\tilde{\boldsymbol{v}}_{ri} \coloneqq [v_{t_1ri}, \ldots, v_{t_Dri}]$, the corresponding entries of $\boldsymbol{V}_t$ across timepoints:

$$\boldsymbol{u}_{rj} \overset{iid}{\sim} N(0, \eta_r^2),\ \tilde{\boldsymbol{v}}_{ri} \overset{iid}{\sim} mvN(0, \boldsymbol{K}_r),\ E_{tji} \overset{iid}{\sim} N(0, \sigma_{tji}^2) \tag{3}$$

where $\boldsymbol{K}_r$ captures component-specific temporal correlation in the factor scores, as detailed in Section 2.4.

2.2. Empirical Variational Bayes Estimation

We begin by describing the estimation algorithm for the rank 1 model, and generalize to the multiple rank scenario in Section 2.3. In a standard empirical Bayes framework, model estimation proceeds by first obtaining point estimates for the prior parameters η^2, $\boldsymbol{K}$, $\boldsymbol{\sigma}^2$, e.g., by maximizing the

marginal log-likelihood under estimated priors $g(\cdot)$:

$$(\hat{\eta}, \hat{\boldsymbol{K}}, \hat{\boldsymbol{\sigma}}^2) = \underset{\boldsymbol{\eta}, \boldsymbol{K}, \boldsymbol{\sigma}^2}{\arg\max} \log \int \int \prod_{t=t_1}^{t_D} p(\boldsymbol{Y}_t | \boldsymbol{u}, \boldsymbol{v}_t, \boldsymbol{\sigma}^2) g(\boldsymbol{v}_t | \boldsymbol{K}) g(\boldsymbol{u} | \boldsymbol{\eta}) d\boldsymbol{v} d\boldsymbol{u}, \tag{4}$$

then performing inference on $\boldsymbol{u}$, $\boldsymbol{v}$ via the posterior distribution

$$p(\boldsymbol{u}, \boldsymbol{v} | \boldsymbol{Y}, \hat{\eta}, \hat{\boldsymbol{K}}, \hat{\boldsymbol{\sigma}}^2). \tag{5}$$

2.2.1. Approximate Inference

The integrals involved in both (4) and (5) are intractable due to marginalizing over latent variables. Therefore, we must resort to approximate inference techniques. Markov Chain Monte Carlo (MCMC) [14] is the most common technique where approximate samples are drawn from the true posterior, however it often suffers from high computational demands for large-scale problems. Instead, we propose a computationally attractive variational Bayes solution for our setting.

In variational Bayes, posterior estimation is posed as an optimization problem, where we search for a factor distribution $q(\boldsymbol{u}, \boldsymbol{v})$ that gives the smallest KL divergence with the true posterior:
$q^*(\boldsymbol{u}, \boldsymbol{v}) = \underset{q \in \mathcal{Q}}{\arg\min}\, \mathrm{KL}(q(\boldsymbol{u}, \boldsymbol{v}) | p(\boldsymbol{u}, \boldsymbol{v} | \boldsymbol{Y}))$ [15]. This is done by maximizing the evidence lower bound (ELBO),

$$\begin{aligned} F_{ELBO}(q, \eta, \boldsymbol{K}, \boldsymbol{\sigma}) &:= E_q[\log \frac{p(\boldsymbol{Y}, \boldsymbol{u}, \boldsymbol{v}; \eta, \boldsymbol{K}, \boldsymbol{\sigma})}{q(\boldsymbol{u}, \boldsymbol{v})}] \\ &= l(\eta, \boldsymbol{K}, \boldsymbol{\sigma}) - \mathrm{KL}(q_{u,v} || p_{u,v|\boldsymbol{Y}, \eta, \boldsymbol{K}, \boldsymbol{\sigma}}) \end{aligned} \tag{6}$$

which is a lower bound for the log marginal likelihood, as the KL divergence is non-negative. Since $l(\eta, \boldsymbol{K}, \boldsymbol{\sigma})$ is constant with respect to $q(\boldsymbol{u}, \boldsymbol{v})$, $\underset{q}{\arg\max}\, F(q, \eta, \boldsymbol{K}, \boldsymbol{\sigma}) = \underset{q}{\arg\min}\, \mathrm{KL}(q_{\boldsymbol{u}, \boldsymbol{v}} || p_{u, \boldsymbol{v}|Y, \eta, \boldsymbol{K}, \boldsymbol{\sigma}})$. However, without further constraints on q, this optimization is difficult. We restrict the space of factor distributions to the mean field variational family,

$$\mathcal{Q} = \{q(\boldsymbol{u}, \boldsymbol{v}) : q(\boldsymbol{u}, \boldsymbol{v}) = \prod_{j=1}^{p} q_{u_j}(u_j) \prod_{t=t_1}^{t_D} \prod_{i=1}^{n} q_{v_{ti}}(v_{ti})\} \tag{7}$$

Under this factorization assumption, we develop a coordinate ascent algorithm to solve for $q^*(\boldsymbol{u}, \boldsymbol{v})$ given by Algorithm 1. To allow for modeling of flexible correlation structure, we define a simpler, more general empirical Bayes model with optimization in the following section that can be used to solve step 4 and step 5 of Algorithm 1.

Algorithm 1 Coordinate Ascent Model Estimation with Rank-1 Model

Require: Initialize $q_u(\boldsymbol{u})^{(0)}$, $q_v(\boldsymbol{v})^{(0)}$, $\eta^{(0)}$, $\boldsymbol{K}^{(0)}$, $\boldsymbol{\sigma}^{(0)}$

1: **repeat**

2: $\quad m \leftarrow m + 1$

3: $\quad \boldsymbol{\sigma}^{(m)} \leftarrow \arg\max_{\boldsymbol{\sigma}} F_{ELBO}(q_{\boldsymbol{u}}^{(m-1)}, q_{\boldsymbol{v}}^{(m-1)}, \eta^{(m-1)}, \boldsymbol{K}^{(m-1)}, \boldsymbol{\sigma})$

4: $\quad q_{\boldsymbol{u}}^{(m)}, \eta^{(m)} \leftarrow \arg\max_{q_{\boldsymbol{u}}, \eta} F_{ELBO}(q_{\boldsymbol{u}}, q_{\boldsymbol{v}}^{(m-1)}, \eta, \boldsymbol{K}^{(m-1)}, \boldsymbol{\sigma}^{(m)})$

5: $\quad q_{\boldsymbol{v}}^{(m)}, \boldsymbol{K}^{(m)} \leftarrow \arg\max_{q_{\boldsymbol{v}}, \boldsymbol{K}} F_{ELBO}(q_{\boldsymbol{u}}^{(m)}, q_{\boldsymbol{v}}, \eta^{(m)}, \boldsymbol{K}, \boldsymbol{\sigma}^{(m)})$

6: **until** convergence criterion met

7: **Return** $q_{\boldsymbol{u}}^{(m)}, q_{\boldsymbol{v}}^{(m)}, \eta^{(m)}, \boldsymbol{K}^{(m)}, \boldsymbol{\sigma}^{(m)}$

2.2.2. Optimizing via Empirical Bayes Random Process Means (EBRPM) Model

In this section, we define the Empirical Bayes Random Process Means Model (EBRPM). This model is closely related to the proposed model, in the sense that there is an equivalence with respect to $\boldsymbol{u}$ or $\boldsymbol{v}$ when the other is fixed. Suppose that we have noisy observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ of true values $\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_n$, each $\boldsymbol{x}_i \in \mathbb{R}^{d_i}$ can potentially be of different dimension:

$$\boldsymbol{x}_i | \boldsymbol{\theta}_i \overset{ind}{\sim} \mathcal{N}_{d_i}(\boldsymbol{\theta}_i, \boldsymbol{S}_i) \tag{8}$$

The $\boldsymbol{S}_i$'s are known diagonal matrices reflecting independent noise and $\boldsymbol{\theta}_i$ are generated from a random process. Here we focus on modeling correlation within the $\boldsymbol{x}_i$'s and restrict our prior family for $\boldsymbol{\theta}_i$ to zero-mean Gaussian processes,

$$\boldsymbol{\theta}_i \overset{iid}{\sim} \mathcal{GP}(0, \boldsymbol{K}(\cdot, \cdot)) \tag{9}$$

since there are many well studied covariance functions and this distribution generally have good computational properties. We are interested in estimating the $\boldsymbol{\theta}_i$'s and $\boldsymbol{K}$ via an empirical Bayes approach, and this is performed

in two steps:

1. Estimate $\boldsymbol{K}$ by type II maximum likelihood:

$$\hat{\boldsymbol{K}} = \arg\max_{\boldsymbol{K}} \int \prod_{i=1}^{n} p(\boldsymbol{x}_i|\boldsymbol{\theta}_i) g(\boldsymbol{\theta}_i|\boldsymbol{K}) d\boldsymbol{\theta} = \arg\max_{\boldsymbol{K}} \prod_{i=1}^{n} \mathcal{N}_{d_i}(\boldsymbol{x}_i|0, \boldsymbol{S}_i + \boldsymbol{K}) \tag{10}$$

2. Calculate the posteriors for $\boldsymbol{\theta_i}$'s using the estimated prior $\hat{\boldsymbol{K}}$ from the first step:

$$p(\boldsymbol{\theta}_i|x, \hat{g}) \propto p(\boldsymbol{x}_i|\boldsymbol{\theta}_i) g(\boldsymbol{\theta}_i|\hat{\boldsymbol{K}}) \tag{11}$$

This model is a special case of the empirical Bayes multivariate normal means model (EBMNM) [16] with multivariate Normal priors when $d_i = d_j$ for all i, j realized at the same timepoints, and the empirical Bayes normal means model (EBNM) [3] with normal priors when $d_i = 1$ for all i. Following the EBNM notation, we summarize the estimation procedure as a function, with an input of $(\boldsymbol{X}, \underset{\sim}{\boldsymbol{S}})$ and output $(p_{\boldsymbol{\theta}|\boldsymbol{x}}, \hat{\boldsymbol{K}})$, where $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n]$, $\underset{\sim}{\boldsymbol{S}} = [\boldsymbol{S}_1, \ldots, \boldsymbol{S}_n]$. In Theorem 2.1 we establish that steps 4 and 5 of Algorithm 1 can be solved by an EBRPM problem.

Theorem 2.1. *Step 4 and 5 of Algorithm 1 are equivalent to the following EBRPM problems:*

$$\arg\max_{q_{\boldsymbol{u}}, \eta} F_{ELBO}(q_{\boldsymbol{u}}, q_{\boldsymbol{v}}, \eta, \boldsymbol{K}, \boldsymbol{\sigma}) = EBRPM(\boldsymbol{x}_j = \frac{\sum_{t,i} Y_{tji} \overline{v}_{ti} / \sigma_t^2}{\sum_{t,i} \overline{v_{ti}^2} / \sigma_t^2}, \boldsymbol{S}_j = \frac{1}{\sum_{t,i} \overline{v_{ti}^2} / \sigma_t^2}) \tag{12}$$

and

$$\arg\max_{q_{\boldsymbol{v}}, \boldsymbol{K}} F_{ELBO}(q_{\boldsymbol{u}}, q_{\boldsymbol{v}}, \eta, \boldsymbol{K}, \boldsymbol{\sigma}) = EBRPM(\boldsymbol{x_i} = \frac{\sum_{j=1}^{p} \overline{u}_j \boldsymbol{y}_{ji}}{\sum_{j=1}^{p} \overline{u_j^2}}, \boldsymbol{S}_i = \frac{1}{\sum_{j=1}^{p} \overline{u_j^2}} \boldsymbol{\Sigma}) \tag{13}$$

Where $\boldsymbol{y}_{ji} = [Y_{t_1 ji}, \ldots, Y_{t_D ji}]^T$, $\boldsymbol{\Sigma} = \mathrm{diag}(\sigma_{t_1}^2, \ldots, \sigma_{t_D}^2)$, $\overline{u}, \overline{v}$ *represent the first and second moments of* $\boldsymbol{u}, \boldsymbol{v}$ *]under* $q_{\boldsymbol{u}}, q_{\boldsymbol{v}}$.

The proof of Theorem 2.1 is provided in Appendix B, and additional computational details are provided in Appendix C and Appendix D.

2.3. Extension to Rank R Model

For the rank R model ($R > 1$), we simply extend the variational approximation by setting factor distributions for all $\boldsymbol{u}_r$, $\boldsymbol{v}_{tr}$ as in (2). F_{ELBO} is then

maximized over $(q_r, \eta_r, \boldsymbol{K}_r)_{r=1}^R, \boldsymbol{\sigma}$ and Algorithm 1 is modified accordingly to iteratively update each factor component while keeping all others constant. The update steps for each $\boldsymbol{u}_r$, $\tilde{\boldsymbol{v}}_{ri}$, $r = 1, \ldots, R$ follows Theorem 2.1, with $\boldsymbol{Y}_t$ replaced by $\boldsymbol{Y}_t - \sum_{r' \neq r} \boldsymbol{u}_{r'} \boldsymbol{v}_{tr'}$. We summarize the estimation prodecure in Algorithm 2.

Algorithm 2 Extension to Rank R Model

Require: Initialize $q_{u_r}(\boldsymbol{u}_r)^{(0)}$, $q_{v_r}(\boldsymbol{v}_r)^{(0)}$, $\eta_r^{(0)}$, $\boldsymbol{K}_r^{(0)}$, $\boldsymbol{\sigma}^{(0)}$, $r = 1, \ldots, R$

repeat
  $m \leftarrow m + 1$
  $\boldsymbol{\sigma}^{(m)} \leftarrow \arg\max_{\boldsymbol{\sigma}} F(q_{\boldsymbol{u}}^{(m-1)}, q_{\boldsymbol{v}}^{(m-1)}, \eta^{(m-1)}, \mathbb{K}^{(m-1)}, \boldsymbol{\sigma})$
  for $r = 1, \ldots, R$ **do**
    $q_{\boldsymbol{u}_r}^{(m)}, \eta_r^{(m)} \leftarrow \arg\max_{q_{\boldsymbol{u}_r}, \eta_r} F(q_{\boldsymbol{u}}, q_{\boldsymbol{v}}^{(m-1)}, \boldsymbol{\eta}, \mathbb{K}^{(m-1)}, \boldsymbol{\sigma}^{(m)})$
    $q_{\boldsymbol{v}_r}^{(m)}, \boldsymbol{K}_r^{(m)} \leftarrow \arg\max_{q_{\boldsymbol{v}_r}, \boldsymbol{K}_r} F(q_{\boldsymbol{u}}^{(m)}, q_{\boldsymbol{v}}, \boldsymbol{\eta}^{(m)}, \mathbb{K}, \boldsymbol{\sigma}^{(m)})$
  end for
until convergence criterion met
Return $q_{\boldsymbol{u}_r}, q_{\boldsymbol{v}_r}, \eta_r, \boldsymbol{K}_r, \boldsymbol{\sigma}, r = 1, \ldots, R$

When R is large, this algorithm can become computationally intensive, as all R factors are updated at each iteration. To alleviate this issue, we implement a greedy algorithm, where rank-1 models are sequentially fitted to the residuals from the sum of previous estimated models, effectively adding one column to the factor matrices at a time, as described in Algorithm 3.

Algorithm 3 Greedy Algorithm for Initialization

Require: Initialize $R = 1$ and $\boldsymbol{S}_t = 0$ for $t = t_1, \ldots, t_D$

repeat
  Set $\boldsymbol{R}_t = \boldsymbol{Y}_t - \boldsymbol{S}_t$ for $t = t_1, \ldots, t_D$
  Perform Algorithm 1 with $\boldsymbol{Y}_t = \boldsymbol{R}_t$ for $t = t_1, \ldots, t_D$
  Set $\hat{\boldsymbol{u}}_R = E_{q_u}[\boldsymbol{U}]$, $\hat{\boldsymbol{v}}_{tR} = E_{q_v}[\boldsymbol{V}_t]$ for $t = t_1, \ldots, t_D$
  Set $\boldsymbol{U} = [\boldsymbol{u}_1 \cdots \boldsymbol{u}_R]$, $\boldsymbol{V}_t = [\boldsymbol{v}_{t1} \cdots \boldsymbol{v}_{tR}]$, $\boldsymbol{S}_t = \boldsymbol{U}\boldsymbol{V}_t^T$ for $t = t_1, \ldots, t_D$
  $R = R + 1$
until convergence criterion met
Return $\boldsymbol{U}, \boldsymbol{V}_t, \boldsymbol{S}_t$ for $t = t_1, \ldots, t_D$

In practice, Algorithm 2 often achieves significantly better fit, given that

it optimizes F_{ELBO} over a larger space. Combining both by using the greedy algorithm as an initialization for the coordinate ascent algorithm results in significantly less iterations required to achieve convergence, thus reducing computational cost without sacrificing model fit.

2.4. Covariance Functions

Several covariance functions for the Gaussian process prior of $\boldsymbol{K}$ have been developed for a wide range of scenarios. We implement several common covariance functions, as follows:

$$\textit{White noise kernel (Indep)}: \quad \boldsymbol{K}(s,t) = \sigma_v^2 \mathbb{1}(s=t) \tag{14}$$

$$\textit{Exchangable covariance (EC)}: \quad \boldsymbol{K}(s,t) = \rho^{\mathbb{1}(s\neq t)}\sigma_v^2 \tag{15}$$

$$\textit{Radial Basis Function kernel (RBF)}: \quad \boldsymbol{K}(s,t) = \sigma_v^2 \exp(-\frac{\|s-t\|^2}{2\alpha^2}) \tag{16}$$

$$\textit{Free covariance (Free)}: \quad \boldsymbol{K}(s,t) = \sigma_{st}^2,\ \boldsymbol{K} \in \mathcal{U}+ \tag{17}$$

where $\mathcal{U}+$ denotes the set of positive definite matrices. We consider the model with independence over time as a benchmark to quantify the performance gain of modeling covariance. This is a special case of the model described in [2] and [4], which has a closed form analytic solution to the empirical variational Bayes approximation. Thus we opt to use their code for faster implementation. Covariance (17) is a free structure with a different covariance for each pair of timepoints. It is impractical at highly irregular timepoints, but is the most flexible structure for the regular timepoint model, and is likely to perform well when the true model structure is unclear. The white noise and exchangeable covariances (14) and (15) have a closed form solution for their parameters in the optimization sub-step (10), while the other covariance structures may be estimated via gradient-based techniques (see Appendix Appendix C for more details). The RBF covariance (16) accounts for distance between timepoints and is well-motivated when observations are irregularly spaced over time.

2.5. Model Dimension Selection

Variational Bayes matrix factorization methods have the advantage of automatically selecting the rank concurrently with the estimation framework [3, 17]. This comes from the fact that F_{ELBO} can be viewed as a balance

between model fit and prior resemblence [15]:

$$F_{ELBO} \propto \sum_{t=t_1}^{t_D} -\frac{1}{2\sigma_t^2} E_q[\|\boldsymbol{Y}_t - \boldsymbol{U}\boldsymbol{V}_t^T\|_F^2] - \mathrm{KL}(q(\boldsymbol{U},\boldsymbol{V})||g(\boldsymbol{U},\boldsymbol{V})) \tag{18}$$

Let $\boldsymbol{U} \in \mathbb{R}^{p\times R_1}, \boldsymbol{V}_t \in \mathbb{R}^{n\times R_1}$ and a larger model $\boldsymbol{U}' \in \mathbb{R}^{p\times R_2}, \boldsymbol{V}_t' \in \mathbb{R}^{n\times R_2}$. It is clear that when $R_2 > R_1$, $\mathrm{KL}(q_{\boldsymbol{u}',\boldsymbol{v}'}||g_{\boldsymbol{u}',\boldsymbol{v}'}) > \mathrm{KL}(q_{\boldsymbol{u},\boldsymbol{v}}||g_{\boldsymbol{u},\boldsymbol{v}})$. So when the decrease in expected squared residuals is less than the penalty incurred by the KL divergence between the added factor distributions and their priors, F_{ELBO} cannot increase and prefers the lower rank model, equivalent to optimizing $q(\boldsymbol{U},\boldsymbol{V})$ towards point masses at zero. Thus, setting R large initially, the backfit algorithm "prunes" the model by setting redundant factors to zero.

Further, rank sparsity is also achieved through the empirical Bayes step, which can minimize the prior density $g(\cdot)$ to a point mass at zero. As noted in [3] and [17], it is possible that the marginal likelihood is maximized for $g(\cdot)$ at zero, and the following lemma from [3] shows that F_{ELBO} is guaranteed to be maximized at $g(\cdot)$ if the marginal likelihood is, which can be applied to our model as well.

Lemma. *For a rank* 1 *model, if* $\arg\max_{g_u,g_v} l(q_{\boldsymbol{u}}, q_{\boldsymbol{v}}, g_{\boldsymbol{u}}, g_{\boldsymbol{v}}, \boldsymbol{\sigma}) = (\delta_{g_u}, \delta_{g_v})$*, then* $\arg\max_{g_u,g_v} F(q_{\boldsymbol{u}}, q_{\boldsymbol{v}}, g_{\boldsymbol{u}}, g_{\boldsymbol{v}}, \boldsymbol{\sigma}) = (\delta_{g_u}, \delta_{g_v})$

In our context, this means that $\boldsymbol{\eta}, \mathbb{K}$ are maximized at 0. This justifies the procedure of the greedy algorithm, as columns will be added to the factor matrices until eventually adding a column of zeros which terminates the algorithm.

2.6. Missing Data Imputation

Here we consider incomplete data and the task of imputing missing elements. Denote the indices of missing entries as $\mathcal{M} = \{(t,j,i) : Y_{tji} \text{ not observed}\}$ and $Y_{\mathcal{M}} = \{Y_{tji} : (t,j,i) \in \mathcal{M}\}$, we propose an Expectation-Maximization type approach, summarized in Algorithm 4.

This approach is analogous to previous methodologies including [4] and [18]. Note that step 4 and 5 are respectively the "maximization" and "expectation" steps, as the former maximizes F_{ELBO} according to Theorem 2.1 and Algorithm 2, and the latter replaces the missing entries with the expected value under current estimates of the factor distributions. This algorithm is

Algorithm 4 Imputation

Set $Y_{tji}^{(0)} = Y_{tji} \cdot \mathbb{1}((t,j,i) \notin \mathcal{M})$
repeat
  $k \leftarrow k+1$
  Perform the complete data algorithm on $\boldsymbol{Y}_t^{(k-1)}$ and denote $\hat{\boldsymbol{S}}_t^{(k)} = E_q[\boldsymbol{U}\boldsymbol{V}_t^T]$ for $t = t_1, \ldots, t_D$
  Impute $\boldsymbol{Y}_{tji}^{(k)} = \hat{\boldsymbol{S}}_{tji}^{(k)}$, $(t,j,i) \in \mathcal{M}$
until $\sum_{(t,j,i)\in\mathcal{M}} (\hat{\boldsymbol{S}}_{tji}^{(k)} - \hat{\boldsymbol{S}}_{tji}^{(k-1)})^2 < \epsilon$ for some small ϵ
Return $\boldsymbol{Y}_{t_1}^{(k)}, \ldots, \boldsymbol{Y}_{t_D}^{(k)}$

justified by Theorem 2.2, which shows the imputation algorithm optimizes F_{EBLO} over missing entries $\boldsymbol{Y}_{\mathcal{M}}$ as well as the latent factor distributions and hyperparameters.

Theorem 2.2. *Algorithm 4 iteratively maximizes* F_{ELBO} *over* $q(\boldsymbol{U}, \boldsymbol{V}), \boldsymbol{\sigma}, \boldsymbol{K}, \boldsymbol{\eta}, \boldsymbol{Y}_{\mathcal{M}}$

The proof of Theorem 2.2 is given in Appendix Appendix B. Compared to the related imputation algorithm in [4], we do not require pre-normalizing the data by estimating $\boldsymbol{\sigma}$ first, but rather incorporate it into the optimization algorithm. This also avoids handling $\boldsymbol{\sigma}$ differently under entrywise and blockwise missingness.

3. Simulations

We conduct two simulation studies to assess both signal recovery and imputation accuracy. We use the regular timepoints model in all simulations to allow for comparisons with competing methods that may not accommodate irregular timepoints.

3.1. Simulation study 1: Complete data signal recovery

Here we assess our model's ability to estimate underlying structure for complete data, and illustrate aspects of the different covariance functions. We generate matrices of size 100×120 for 5 regularly spaced timepoints with true rank 3 according to our model description, under each of *EC*, *RBF*, and *Free* covariance structures with varying signal-to-noise ratios (SNR) and correlation levels. We also use a block covariance structure that none of the covariance functions correctly capture to assess misspecification, where

an EC correlation is given independently for different subsets of timepoints. See Appendix E for complete details on the data generation for each scenario. We implement our model using the three non-trivial covariance functions in Section 2.4 (EC, RBF, Free), and without correlation using the EVB implementation in [4]. We also consider the general tensor decomposition method TEMPTED that doesn't explicitly model covariance between timepoints but rather treats time as a continuous variable [9]. Accuracy is measured using the relative squared error:

$$\text{RSE} = \sum_{t,j,i} (\hat{S}_{tji} - S_{tji})^2 / \sum_{t,j,i} S_{tji}^2 \tag{19}$$

where S_{tji} is the low-rank signal: $\mathbf{S}_t = \mathbf{U}\mathbf{V}^T$. Table 1 shows the results for all methods. For each covariance structure, the correctly specified model consistently performs the best, as one would reasonably expect. When SNR is high, the model can perform well without modeling correlation, even when the true underlying structure is highly correlated. However, the margin of improvement in estimation accuracy is more significant under low SNR settings, demonstrating that the borrowing of information is effective when the data itself has low information. The TEMPTED method does not perform in a comparable range in this experiment as it does not assume a correlated low rank matrix structure. In the misspecified scenario, the *free* covariance model performs consistently the best across SNR ratios and correlation level, since it can be considered as correctly specified but overparametrized, which leads to more robust estimation at the cost of lower efficiency. Lastly, we note that in all simulation repetitions, all models selected the true rank.

3.2. Simulation study 2: Missing data imputation

Here we evaluate the imputation performance of our model. We consider both entrywise and columnwise missingness, where the latter refers to entire observations missing at a timepoint. Data are generated as in Section 3.1, then 20%, 30%, and 50% of the data are randomly set to be missing. In addition to the methods in experiment 1, we compare our model against two imputation algorithms - BAMITA, which is a Bayesian tensor imputation model [10], as well as MissForest, a non-parametric, random forest-based imputation method that doesn't assume low rank latent structure [19]. We evaluate imputation accuracy by restricting to the RSE for missing entries

Table 1: RSE for complete data estimation

RBF Scenario						
	SNR = 0.2		SNR = 0.5		SNR = 1	
Method	Low Corr	High Corr	Low Corr	High Corr	Low Corr	High Corr
EC	0.239 (0.007)	0.184 (0.005)	0.042 (0.001)	0.043 (0.001)	0.012 (0.000)	0.011 (0.000)
RBF	**0.213** (0.006)	**0.139** (0.005)	**0.040** (0.001)	**0.029** (0.001)	**0.011** (0.000)	**0.008** (0.000)
Free	0.218 (0.006)	0.149 (0.005)	0.040 (0.001)	0.030 (0.001)	**0.011** (0.000)	0.008 (0.000)
EVB	0.251 (0.007)	0.251 (0.008)	0.042 (0.001)	0.047 (0.002)	0.012 (0.000)	0.012 (0.000)
TEMPTED	0.509 (0.023)	0.515 (0.018)	0.110 (0.002)	0.080 (0.003)	0.073 (0.001)	0.016 (0.001)
EC Scenario						
EC	**0.244** (0.007)	**0.181** (0.003)	**0.045** (0.001)	**0.040** (0.001)	**0.012** (0.000)	**0.011** (0.000)
RBF	0.248 (0.007)	0.231 (0.003)	**0.045** (0.001)	0.042 (0.001)	**0.012** (0.000)	**0.011** (0.000)
Free	0.247 (0.007)	0.189 (0.003)	**0.045** (0.001)	**0.040** (0.001)	**0.012** (0.000)	**0.011** (0.000)
EVB	0.254 (0.007)	0.251 (0.005)	**0.045** (0.001)	0.043 (0.001)	**0.012** (0.000)	0.012 (0.000)
TEMPTED	0.647 (0.018)	0.639 (0.017)	0.247 (0.004)	0.109 (0.003)	0.208 (0.003)	0.062 (0.002)
Free Scenario						
EC	0.243 (0.024)		0.046 (0.005)		0.015 (0.001)	
RBF	0.242 (0.024)		0.046 (0.005)		0.015 (0.001)	
Free	**0.203** (0.022)		**0.040** (0.005)		**0.013** (0.001)	
EVB	0.248 (0.026)		0.047 (0.006)		0.015 (0.001)	
TEMPTED	0.536 (0.080)		0.089 (0.009)		0.050 (0.006)	
Block Scenario						
EC	**0.250** (0.006)	0.232 (0.004)	**0.045** (0.001)	0.044 (0.001)	**0.012** (0.000)	**0.011** (0.000)
RBF	**0.250** (0.006)	0.225 (0.004)	**0.045** (0.001)	0.043 (0.001)	**0.012** (0.000)	**0.011** (0.000)
Free	0.252 (0.006)	**0.203** (0.004)	**0.045** (0.001)	**0.042** (0.001)	**0.012** (0.000)	**0.011** (0.000)
EVB	0.256 (0.006)	0.245 (0.005)	**0.045** (0.001)	0.044 (0.001)	**0.012** (0.000)	**0.011** (0.000)
TEMPTED	0.669 (0.018)	0.551 (0.012)	0.260 (0.003)	0.111 (0.002)	0.227 (0.003)	0.068 (0.002)

only.

$$\text{RSE}_{miss} = \sum_{(t,j,i)\in\mathcal{M}} (\hat{S}_{tji} - S_{tji})^2 / \sum_{(t,j,i)\in\mathcal{M}} S_{tji}^2 \tag{20}$$

We note that TEMPTED cannot handle entrywise missing data so it is excluded from the setting. The results for entrywise missingness are shown in Table 2. The patterns closely mirror the complete data estimation results from Table 1, where the correctly specified covariance model consistently performs the best, especially in low SNR settings. Alternative imputation methods perform markedly worse across all scenarios, with BAMITA performing substantially worse in high SNR settings, due to the strong signal indicating a non-tensor data structure. MissForest exhibits substantially higher errors potentially due to not assuming low rank structure or an explicit correlation structure across time.

Table 3 summarizes the results for columnwise missingness imputation. The task becomes much more difficult, as imputation completely relies on shared information between timepoints. This is shown by the results of low

Table 2: RSE for entrywise missingness imputation

RBF Scenario						
	SNR = 0.2		SNR = 0.5		SNR = 1	
Method	Low Corr	High Corr	Low Corr	High Corr	Low Corr	High Corr
EC	0.280 (0.010)	0.212 (0.007)	0.051 (0.001)	0.049 (0.002)	**0.013** (0.000)	0.013 (0.001)
RBF	**0.249** (0.009)	**0.161** (0.006)	**0.048** (0.001)	**0.033** (0.001)	**0.013** (0.000)	**0.009** (0.000)
Free	0.255 (0.009)	0.187 (0.009)	0.049 (0.001)	0.034 (0.001)	**0.013** (0.000)	**0.009** (0.000)
EVB	0.295 (0.010)	0.295 (0.008)	0.052 (0.001)	0.054 (0.002)	0.014 (0.000)	0.013 (0.001)
BAMITA	0.390 (0.012)	0.272 (0.009)	0.115 (0.002)	0.053 (0.002)	0.075 (0.001)	0.013 (0.001)
MissForest	0.802 (0.018)	0.810 (0.015)	0.211 (0.004)	0.223 (0.004)	0.089 (0.002)	0.089 (0.003)
EC Scenario						
	SNR = 0.2		SNR = 0.5		SNR = 1	
Method	Low Corr	High Corr	Low Corr	High Corr	Low Corr	High Corr
EC	**0.287** (0.012)	**0.199** (0.006)	**0.051** (0.002)	**0.048** (0.001)	**0.014** (0.000)	**0.013** (0.000)
RBF	0.290 (0.013)	0.246 (0.006)	**0.051** (0.002)	0.051 (0.002)	**0.014** (0.000)	0.014 (0.000)
Free	0.291 (0.012)	0.211 (0.007)	**0.051** (0.002)	**0.048** (0.002)	**0.014** (0.000)	**0.013** (0.000)
EVB	0.296 (0.013)	0.291 (0.009)	0.052 (0.002)	0.053 (0.002)	**0.014** (0.000)	0.014 (0.000)
BAMITA	0.528 (0.013)	0.314 (0.008)	0.240 (0.004)	0.103 (0.002)	0.206 (0.005)	0.065 (0.002)
MissForest	0.800 (0.019)	0.804 (0.020)	0.209 (0.006)	0.210 (0.006)	0.088 (0.003)	0.088 (0.003)
Free Scenario						
	SNR = 0.2		SNR = 0.5		SNR = 1	
Method	Single Condition					
EC	0.332 (0.027)		0.051 (0.004)		0.015 (0.001)	
RBF	0.331 (0.028)		0.051 (0.004)		0.015 (0.001)	
Free	**0.283** (0.040)		**0.043** (0.004)		**0.014** (0.001)	
EVB	0.339 (0.029)		0.051 (0.004)		0.015 (0.001)	
BAMITA	0.401 (0.040)		0.082 (0.007)		0.051 (0.006)	
MissForest	0.883 (0.052)		0.215 (0.014)		0.098 (0.005)	
Block Scenario						
	SNR = 0.2		SNR = 0.5		SNR = 1	
Method	Low Corr	High Corr	Low Corr	High Corr	Low Corr	High Corr
EC	**0.286** (0.005)	0.277 (0.007)	**0.051** (0.001)	0.054 (0.001)	**0.013** (0.000)	0.014 (0.000)
RBF	0.287 (0.005)	0.266 (0.006)	**0.051** (0.001)	0.053 (0.001)	**0.013** (0.000)	0.014 (0.000)
Free	0.288 (0.005)	**0.245** (0.006)	**0.051** (0.002)	**0.050** (0.001)	**0.013** (0.000)	**0.013** (0.000)
EVB	0.291 (0.005)	0.295 (0.008)	**0.051** (0.001)	0.054 (0.001)	**0.013** (0.000)	0.014 (0.000)
BAMITA	0.517 (0.007)	0.363 (0.009)	0.255 (0.005)	0.111 (0.002)	0.212 (0.004)	0.066 (0.001)
MissForest	0.795 (0.010)	0.800 (0.014)	0.209 (0.005)	0.222 (0.004)	0.086 (0.003)	0.090 (0.002)

correlation scenarios under both the EC and RBF correlation structure, where RSE vary little over SNR levels. Even for the correct model, RSE is significantly worse than in estimation or entrywise imputation. All alternative methods other than BAMITA have RSE$>$ 1, which is the RSE of null imputing with 0. It is also worth noting that misspecification of the correlation structure is more costly, as shown by the relatively worse performance of our model in misspecified scenarios compared to estimation and entrywise imputation. Only the most flexible free covariance could still achieve an RSE well below 1. On the other hand, it is remarkable that the proposed method performs relatively well even under the lowest SNR under high correlation, recovering about 70% of true structure in both the EC and RBF scenarios.

Table 3: RSE of blockwise imputation for all methods under varying simulated conditions, with the best performing method highlighted in bold font.

RBF Scenario						
	SNR = 0.2		SNR = 0.5		SNR = 1	
Method	Low Corr	High Corr	Low Corr	High Corr	Low Corr	High Corr
EC	0.880 (0.071)	0.352 (0.037)	0.843 (0.057)	0.291 (0.052)	0.847 (0.022)	0.282 (0.038)
RBF	**0.627** (0.061)	**0.234** (0.033)	**0.462** (0.050)	**0.083** (0.014)	**0.449** (0.063)	**0.042** (0.009)
Free	0.670 (0.066)	0.264 (0.042)	0.476 (0.050)	0.089 (0.015)	0.462 (0.068)	0.044 (0.008)
EVB	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)
BAMITA	0.874 (0.082)	0.350 (0.061)	0.652 (0.105)	0.124 (0.021)	0.900 (0.233)	0.059 (0.013)
MissForest	11.205 (1.150)	10.368 (1.577)	3.430 (0.432)	3.392 (0.346)	2.443 (0.319)	2.279 (0.219)
TEMPTED	2.005 (0.401)	1.978 (0.132)	1.643 (0.204)	1.776 (0.197)	1.508 (0.139)	1.908 (0.300)
EC Scenario						
EC	**0.928** (0.030)	**0.335** (0.027)	**0.931** (0.036)	**0.278** (0.033)	**0.916** (0.018)	**0.254** (0.027)
RBF	0.976 (0.018)	0.380 (0.048)	0.973 (0.028)	0.557 (0.064)	0.970 (0.020)	0.506 (0.050)
Free	0.964 (0.028)	0.373 (0.028)	0.975 (0.064)	0.294 (0.039)	0.932 (0.025)	0.263 (0.032)
EVB	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)
BAMITA	1.325 (0.163)	0.451 (0.033)	2.723 (0.255)	0.516 (0.069)	5.835 (1.340)	0.812 (0.220)
MissForest	9.877 (1.417)	10.730 (1.043)	3.556 (0.396)	3.282 (0.224)	2.459 (0.164)	2.249 (0.213)
TEMPTED	1.723 (0.216)	2.172 (0.317)	1.519 (0.140)	1.811 (0.188)	1.500 (0.162)	1.856 (0.273)
Free Scenario						
EC	0.999 (0.014)		0.970 (0.042)		0.986 (0.032)	
RBF	0.981 (0.037)		0.971 (0.035)		0.976 (0.040)	
Free	**0.662** (0.196)		**0.364** (0.126)		**0.318** (0.133)	
EVB	1.000 (0.000)		1.000 (0.000)		1.000 (0.000)	
BAMITA	0.807 (0.222)		0.477 (0.147)		0.673 (0.415)	
MissForest	11.321 (2.702)		3.653 (0.504)		2.353 (0.165)	
TEMPTED	2.041 (0.358)		1.793 (0.183)		1.784 (0.234)	
Block Scenario						
EC	**0.979** (0.029)	0.866 (0.071)	0.981 (0.018)	0.843 (0.054)	0.991 (0.020)	0.868 (0.028)
RBF	0.983 (0.020)	0.697 (0.063)	**0.972** (0.023)	0.651 (0.058)	0.988 (0.015)	0.690 (0.044)
Free	0.999 (0.045)	**0.549** (0.041)	0.990 (0.047)	**0.405** (0.040)	**0.969** (0.030)	**0.418** (0.056)
EVB	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)	1.000 (0.000)
BAMITA	1.416 (0.123)	0.700 (0.053)	3.010 (0.381)	0.633 (0.071)	5.545 (1.837)	0.985 (0.125)
MissForest	10.182 (0.997)	11.127 (1.703)	3.275 (0.332)	3.389 (0.325)	2.384 (0.167)	2.209 (0.134)
TEMPTED	1.801 (0.163)	2.093 (0.332)	1.616 (0.155)	1.686 (0.241)	1.564 (0.112)	26.49 (77.14)

4. Gene Time Course Data Application

We apply our method to a set of gene time course data described in [6]. This study explored methods to classify genetic response of multiple sclerosis patients to recombinant human interferon beta(rIFNβ) treatment. rIFNβ was given to 53 patients and expression levels of 76 genes were measured at 7 timepoints (baseline and follow-ups at $3, 6, 9, 12, 18, 24$ months after treatment). Most patients had some percentage of missingness, ranging from $1-6$ entire timepoints missing and additional partial missingness at observed timepoints. Overall, 10.5% of all data were missing. We first use this dataset to evaluate the empirical imputation performance of our model. Similar to the experiments in section 3.2, we apply our model with the three non-trival covariance functions, EVB under independence, BAMITA, and MissForest to the data. TEMPTED is omitted as it cannot handle entrywise missingness. We take the complete indices and perform $10, 3$, 2-fold cross validation based on the missing proportion, repeated 10 times over different random seeds. Figure 1 shows the results.

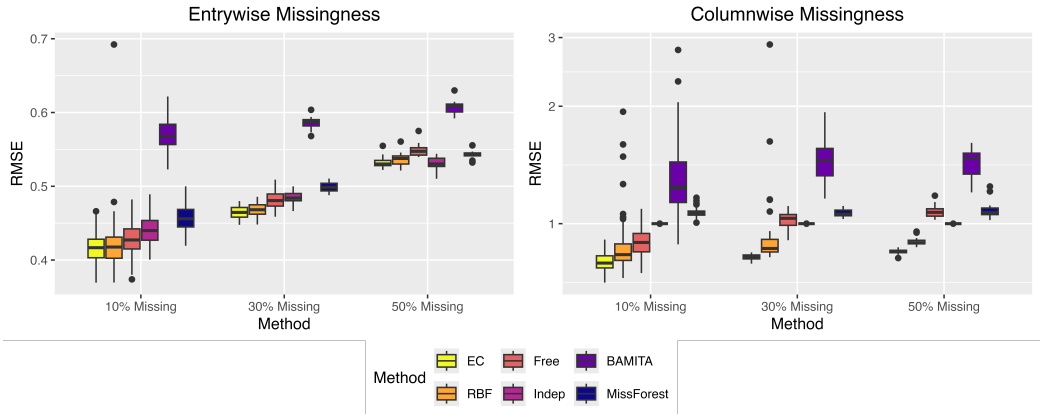
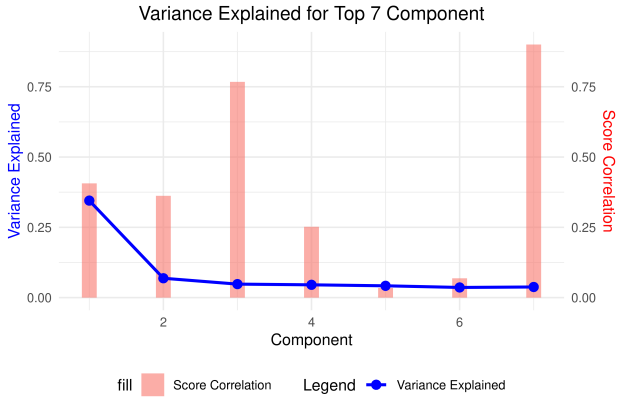
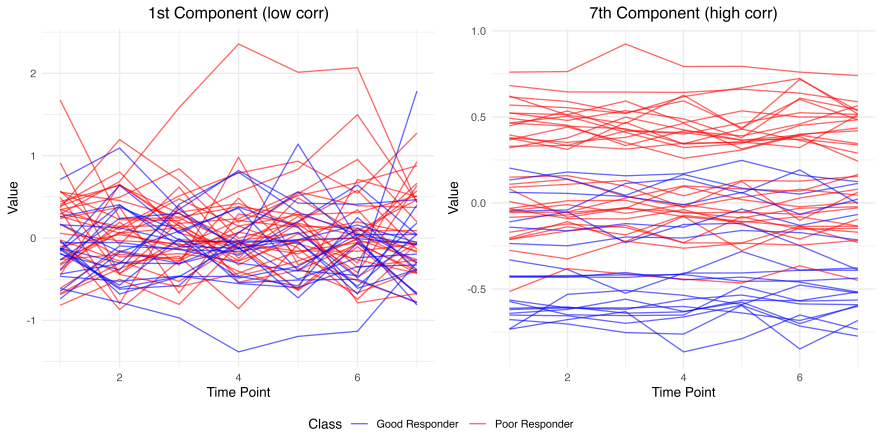


Figure 1: Analysis results for the IFNβ data application: Imputation accuracy for all methods from cross validation.

In the comparisons, the *EC* covariance model performed the best across both entrywise and columnwise missingness. The independent model performed comparably in the entrywise missing scenarios, but had $RSE = 1$ in columnwise missingness, consistent with simulation results. All alternative methods performed worse than the independent model, highlighting the advantage of our proposed model in columnwise imputation.



(a) Variance explained and correlation for the first 7 components.



(b) Individual trajectories for the first and seventh component scores.

Figure 2: Analysis results for the IFNβ data application.

As the *EC* covariance model performed the best, we refer to the results produced by this model for interpretation. We estimated a total of 14 components; the top 7 ordered by variance explained are shown in Figure 2a, along with the estimated correlation in each component's scores. The first component explains about 35% variance, while the next 6 each explains less than 10% variance each. Both the third and seventh component exhibits high correlation between sample scores at different timepoints. We plot individual score trajectories in Figure 2b, colored by manually labeled good/poor

responders from the original study. The first component, which is the major contributor, exhibits similar patterns for most individuals, indicating a mean effect of subjects. The seventh component, on the other hand, shows a clear clustering effect in good and poor responders, with good responders at the bottom and poor responders at the top. The individual scores have little variation between time points, which indicates that the two types of responders can be distinguished without inferring from temporal patterns. This is consistent with the results in [6], where no group*time interaction effects were discovered; as well as [20], where distinction between the two groups were not driven by factors that vary over time. More generally, the EC covariance and accompanying score trajectories suggest some clear subject effects but no other temporal trends.

5. Discussion

In this paper, we introduce a latent matrix factorization method for high dimensional longitudinal data that captures temporal correlation by flexibly modeling the covariance structure in latent factor scores. The additive nature of our model enables decomposition of data into static and temporal factors, enhancing interpretability. The model employs a computationally efficient empirical variational Bayes estimation algorithm that automatically estimates hyperparameters and selects model dimension by optimizing an objective function balancing model fit and complexity. Through extensive simulation studies and a real world data application, the proposed method demonstrates superior estimation and imputation accuracy compared to existing approaches across diverse scenarios.

While we focus on single-source data, multimodal datasets combining data from multiple sites are increasingly common in biomedical studies, such as collaborative neurodegenerative disease research that collects imaging, clinical, and genetic modalities [21, 22, 23]. Extensions of our method to multimodal matrix factorization—where some components are restricted to specific cohorts or data modalities—is an interesting direction for future research, as the full rank R and greedy estimation algorithms do not straightforwardly extend to such settings.

A known limitation of variational Bayes methods is uncertainty quantification. While point estimates generally perform well and have good properties, variational Bayes methods typically underestimate marginal posterior variances [24, 25]. To address this limitation, our method can serve as initial-

ization for the associated fully Bayesian model, offering several advantages including initialization near the posterior mode and automatic rank selection, which are challenging for the fully Bayesian approach to achieve alone. A related Bayesian model is described in [26], which illustrates the benefits of accounting for uncertainty in imputation and prediction. However, standard MCMC strategies such as Gibbs sampling would likely face substantial computational challenges, necessitating the development of more efficient sampling algorithms.

Code & Data Availability

Documented R code to perform the proposed method and the IFNβ dataset analyzed in Section 4 is available at
https://github.com/andyljk/Longitudinal_Factorization.

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Appendix A. Irregular Timepoint Model

In this section, we provide a description of the key differences of the extended model that handles inconsistency in measured timepoints across individuals. For the $i = 1, \ldots, n$ individual, assume each with p variables measured at time $t_1^{(i)}, \ldots, t_{d_i}^{(i)}$. The outcome for individual i, variable j at time t is modeled as

$$\boldsymbol{Y}_{tji} = \sum_{r=1}^{R} \boldsymbol{u}_{rj} \boldsymbol{v}_{tri} + \boldsymbol{E}_{tji} \tag{A.1}$$

Note that the matrix product representation in (1) is not available in this setting, since $\boldsymbol{v}$ does not have consistent measures at each timepoint. Priors are prescribed to each of $\boldsymbol{u}, \boldsymbol{v}$ similar to (3), where

$$\boldsymbol{u}_{rj} \overset{iid}{\sim} N(0, \eta_r^2),\ \tilde{\boldsymbol{v}}_{ri} \overset{iid}{\sim} mvN(0, \boldsymbol{K}_r^{(i)}),\ e_{tji} \overset{iid}{\sim} N(0, \sigma_{tji}^2) \tag{A.2}$$

and $\tilde{\boldsymbol{v}}_{ri} = [\boldsymbol{v}_{t_1^{(i)} ri} \cdots \boldsymbol{v}_{t_{d_i}^{(i)} ri}] \in \mathbb{R}^{d_i}$. The prior covariance matrix $\boldsymbol{K}_r^{(i)} \in \mathbb{R}^{d_i \times d_i}$ has dimension matching the number of measurements from individual i. We assume all prior covariance are parametrized by a common set of parameters as described in Section 2.4, and denote estimates of $\boldsymbol{K}$ as $\hat{\boldsymbol{K}}$ for convenience. When the measurements are highly irregular, modeling the noise variance separately at each timepoints becomes difficult. In this situation, a reasonable modification is to set $\sigma_t^2 = \sigma^2$ for all t.

Denoting $\boldsymbol{U} = \{\boldsymbol{u}_{jr} : j = 1, \ldots, p; r = 1, \ldots, R\}$ and $\boldsymbol{V} = \{\boldsymbol{v}_{tir} : t \in T; i = 1, \ldots, n; r = 1, \ldots, R\}$, we maximize the same objective function $F_{ELBO}(q, \boldsymbol{\eta}, \boldsymbol{K}, \sigma)$ using algorithm 2. For the rank 1 model, it can be shown following the proof of Theorem 2.1, step 4, 5 of algorithm 1 are equivalent to solving the following

$$\arg\max_{q_{\boldsymbol{u}},\eta} F_{ELBO}(q_{\boldsymbol{u}}, q_{\boldsymbol{v}}, \eta, \boldsymbol{K}, \boldsymbol{\sigma}) = \text{EBRPM}(\boldsymbol{x}_j = \frac{\sum_{i=1}^{n}\sum_{t=t_1^{(i)}}^{t_{d_i}^{(i)}} Y_{tji}\overline{v}_{ti}/\sigma_t^2}{\sum_{i=1}^{n}\sum_{t=t_1^{(i)}}^{t_{d_i}^{(i)}} \overline{v_{ti}^2}/\sigma_t^2}, \boldsymbol{S}_j = \frac{1}{\sum_{i=1}^{n}\sum_{t=t_1^{(i)}}^{t_{d_i}^{(i)}} \overline{v_{ti}^2}/\sigma_t^2}) \tag{A.3}$$

and

$$\arg\max_{q_{\boldsymbol{v}},\boldsymbol{K}} F_{ELBO}(q_{\boldsymbol{u}}, q_{\boldsymbol{v}}, \eta, \boldsymbol{K}, \boldsymbol{\sigma}) = \text{EBRPM}(\boldsymbol{x_i} = \frac{\sum_{j=1}^{p}\overline{u}_j \boldsymbol{y}_{ji}}{\sum_{j=1}^{p}\overline{u_j^2}}, \boldsymbol{S}_i = \frac{1}{\sum_{j=1}^{p}\overline{u_j^2}}\boldsymbol{\Sigma_i}) \tag{A.4}$$

Note that each $\boldsymbol{y}_{ji}$ is of potentially different dimension, which fits in the framework of the EBRPM model by design. For estimation of the multiple rank model, steps $5, 6$ of algorithm 2 and step 3 of algorithm 3 are modified accordingly to accomodate the timepoint structure.

Appendix B. Proofs

Appendix B.1. Theorem 2.1

We prove Theorem 2.1, where we stated the equivalence between step 4, 5 of algorithm 1 and the EBRPM problem. The proof is similar to Proposition 5 in [3], which can be viewed as a univariate case of our model. We first introduce the following lemma as an intermediate step.

Lemma. *For $i = 1, \ldots, n$, let $\boldsymbol{x}_i \overset{iid}{\sim} mvN_{d_i}(\boldsymbol{\theta}_i, \boldsymbol{S}_i)$, $\boldsymbol{\theta}_i \overset{iid}{\sim} \mathcal{GP}(0, \boldsymbol{K})$, and $\mathbb{S} = \{\boldsymbol{S}_i : i = 1, \ldots, n\}$ be known. Recall the EBRPM first estimates the prior variance function as $\hat{\boldsymbol{K}} = \arg\max_{\boldsymbol{K}} \log p(\boldsymbol{x}|\boldsymbol{K})$, then uses this estimate to calculate the posterior $p(\boldsymbol{\theta}|\boldsymbol{x}, \hat{\boldsymbol{K}})$. The solution to the EBRPM problem $(p(\boldsymbol{\theta}|\boldsymbol{x}, \hat{\boldsymbol{K}}), \hat{\boldsymbol{K}})$ is equivalent to the solution of the following objective:*

$$\arg\max_{q,\boldsymbol{K}} F_0(\boldsymbol{K}, q(\boldsymbol{\theta})) \tag{B.1}$$

where

$$
\begin{aligned}
F_0(\boldsymbol{K}, q(\boldsymbol{\theta})) &= E_q[\log \frac{p(\boldsymbol{x}, \boldsymbol{\theta}|K, \mathbb{S})}{q(\boldsymbol{\theta})}] \\
&\propto E_q[-1/2 \sum_{i=1}^{n} \boldsymbol{\theta}_i^T A_i \boldsymbol{\theta}_i - 2B_i^T \boldsymbol{\theta}_i] + E_q[\log \frac{g(\boldsymbol{\theta}|K)}{q(\boldsymbol{\theta})}]
\end{aligned}
\tag{B.2}
$$

$$
\textit{where } \boldsymbol{A}_i = \boldsymbol{S}_i^{-1},\ \boldsymbol{B}_i = S_i^{-1}\boldsymbol{x}_i
$$

Proof. To demonstrate equivalence, we need to show that $\hat{q}(\boldsymbol{\theta}) = p(\boldsymbol{\theta}|\boldsymbol{x}, \hat{K})$ and $\arg\max_K F_0 = \hat{K}_{EB}$. Note that

$$
\begin{aligned}
\log p(\boldsymbol{x}|K) &= \int q(\boldsymbol{\theta}) \log \frac{p(\boldsymbol{x}, \boldsymbol{\theta}|K, \boldsymbol{S})}{q(\boldsymbol{\theta})} d\boldsymbol{\theta} + \int q(\boldsymbol{\theta}) \log \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta}|\boldsymbol{x}, K)} d\boldsymbol{\theta} \\
&= F_0(K, q(\boldsymbol{\theta})) + KL(q_\theta || p_{\theta|x})
\end{aligned}
\tag{B.3}
$$

Since $KL(q_\theta||p_{\theta|x}) \geq 0$ with equality iff $q_\theta = p_{\theta|x}$, we see that

$$
\max_q F_0(q(\boldsymbol{\theta}), K) = \log p(\boldsymbol{x}|K) \tag{B.4}
$$

when $q(\boldsymbol{\theta}) = p(\boldsymbol{\theta}|\boldsymbol{x}, K)$, so then

$$
\arg\max_K \{\max_q F_0(q(\boldsymbol{\theta}), K)\} = \arg\max_K \log p(\boldsymbol{x}|K) = \hat{K}_{EB} \tag{B.5}
$$

Lastly we show that F_0 has the expanded form, which can be done by straightforwardly factorizing the joint distribution:

$$
\begin{aligned}
F_0(K, q(\boldsymbol{\theta})) &= E_q[\log \frac{p(\boldsymbol{x}, \boldsymbol{\theta}|K, \boldsymbol{S})}{q(\boldsymbol{\theta})}] \\
&= E_q[\log p(\boldsymbol{x}|\boldsymbol{\theta}, K, \boldsymbol{S})] + E_q[\log \frac{g(\boldsymbol{\theta}|K)}{q(\boldsymbol{\theta})}] \\
&= E_q[\sum_{i=1}^{n} \log N_{d_i}(\boldsymbol{x}_i|\boldsymbol{\theta}_i, S_i)] + E_q[\log \frac{g(\boldsymbol{\theta}|K)}{q(\boldsymbol{\theta})}] \\
&\propto E_q[-1/2 \sum_{i=1}^{n} \boldsymbol{\theta}_i^T A_i \boldsymbol{\theta}_i - 2B_i^T \boldsymbol{\theta}_i] + E_q[\log \frac{g(\boldsymbol{\theta}|K)}{q(\boldsymbol{\theta})}]
\end{aligned}
\tag{B.6}
$$

□

We now prove Thoerem 2.1.

Proof. In step 4 of Algorithm 1, we treat all quantities other than $q_{\boldsymbol{u}}, \eta$ as fixed. The objective function is then

$$\begin{aligned} F(q, \eta, K, \boldsymbol{\sigma}) &:= E_q[\log \frac{p(Y, u, v; \eta, K, \boldsymbol{\sigma})}{q(u, v)}] \\ &\propto E_{q_{\boldsymbol{u}}}[\sum_{t=t_1}^{t_D} \log p(Y_t | \boldsymbol{u}, \boldsymbol{v}_t, \eta, K)] + E_{q_{\boldsymbol{u}}}[\log \frac{g(\boldsymbol{u}|\eta)}{q(\boldsymbol{u})}] \\ &\propto E_{q_{\boldsymbol{u}}}[-1/2 \sum_{j=1}^{p} \boldsymbol{u}_j^T A_j \boldsymbol{u}_j - 2B_j^T \boldsymbol{u}_j] + E_{q_{\boldsymbol{u}}}[\log \frac{g(\boldsymbol{u}|\eta)}{q(\boldsymbol{u})}] \end{aligned} \tag{B.7}$$

$$\text{Where } A_j = \sum_{t=t_1}^{t_D} \sum_{i=1}^{n} \overline{v^2}_{ti}/\sigma_t^2, \; B_j = \sum_{t=t_1}^{t_D} \sum_{i=1}^{n} Y_{tji} \overline{v}_{ti}/\sigma_t^2$$

Then equation 12 follows from Lemma Appendix B.1. Similarly, with respect to $q_{\boldsymbol{v}}, K$ for step 5, the objective function becomes

$$F(q, \eta, K, \boldsymbol{\sigma}) \propto E_{q_{\boldsymbol{v}}}[-1/2 \sum_{i=1}^{n} \boldsymbol{v}_i^T A_i \boldsymbol{v}_i - 2B_i^T \boldsymbol{v}_i] + E_{q_{\boldsymbol{v}}}[\log \frac{g(\boldsymbol{v}|K)}{q(\boldsymbol{v})}] \tag{B.8}$$

$$\text{Where } A_i = (\sum_{j=1}^{p} \overline{u_j^2}) \boldsymbol{\Sigma}^{-1}, \; B_i = \boldsymbol{\Sigma}^{-1} (\sum_{j=1}^{p} \overline{u}_j \boldsymbol{y}_{ji}), \; \boldsymbol{\Sigma} = \text{diag}(\sigma_{t_1}^2, \ldots, \sigma_{t_D}^2)$$

and equation 13 follows from Lemma Appendix B.1. □

Appendix B.2. Theorem 2.2

Proof. From algorithm 2, it can be seen that the M steps of algorithm 4 each partially maximize the ELBO over $\boldsymbol{\sigma}, q(U), q(\mathbb{V})$, holding all other quantities fixed. So it leaves to show that the E step maximizes the ELBO over the missing entries $Y_{\mathcal{M}}$. The ELBO can be decomposed into

$$F_{ELBO} = E_q[\sum_{t=t_1}^{t_D} \log p(Y_t | U, V_t, \boldsymbol{\eta}, \mathbb{K})] + E_{q_{\boldsymbol{u}}}[\log \frac{g(U|\boldsymbol{\eta})}{q(U)}] + E_{q_{\boldsymbol{v}}}[\log \frac{g(\mathbb{V}|\mathbb{K})}{q(\mathbb{V})}] \tag{B.9}$$

holding all other quantities fixed, F_{ELBO} with respect to $Y_{\mathcal{M}}$ is proportional to

$$\begin{aligned} F_{ELBO}(Y_{\mathcal{M}}) &\propto E_q[\sum_{(t,j,i)\in\mathcal{M}} \log p(Y_{tji}|\boldsymbol{u}_j, \boldsymbol{v}_{ti}, \boldsymbol{\eta}, \mathbb{K})] \\ &= \sum_{(t,j,i)\in\mathcal{M}} -\frac{1}{2\sigma_t^2}(Y_{tji} - \sum_{r=1}^{R} E_q[U_{jr}V_{tir}])^2 + b \leq b \\ &\text{where } b \text{ is a constant} \end{aligned} \tag{B.10}$$

Thus it can be seen that F_{ELBO} is maximized at $Y_{tji} = \sum_{r=1}^{R} E_q[U_{jr}V_{tir}] = \hat{S}_{tji}$, for all $(t, j, i) \in \mathcal{M}$. □

Appendix C. Solving the EBRPM problem

Here, we provide the details for implementing the two step solution to solve the EBRPM problem described in Section 2.2.2. From the model described in equation 8, 9, the first step is to estimate K by maximizing the marginal likelihood. We restrict to the case where K is parametrized by some parameters $\boldsymbol{\alpha}$, and estimate α instead, which is the case that all covariance functions discussed fall into. Below, we calculate the analytic form of the gradients and several well-documented gradient-based optimization methods can be used. For convenience, let $K_i = S_i + K$, the marginal log likelihood is given by:

$$l(\boldsymbol{\alpha}) = -\frac{1}{2}\sum_{i=1}^{n}\{d_i \log 2\pi + \sum_{i=1}^{n} \log|K_i| + \boldsymbol{x}_i^T K_i^{-1} \boldsymbol{x}_i\} \tag{C.1}$$

It can be shown that the matrix derivative of inverse matrices and log determinant of positive definite matrices are given by

$$\frac{\partial}{\partial\alpha} K_i^{-1} = -K_i^{-1}\frac{\partial K_i}{\partial\alpha}K_i^{-1} \tag{C.2}$$

$$\frac{\partial}{\partial\alpha}\log|K_i| = \mathrm{tr}(K_i^{-1})\frac{\partial K_i}{\partial\alpha} \tag{C.3}$$

Where $\frac{\partial K_i}{\partial\alpha}$ denotes the elementwise derivative matrix. Thus the gradient is given by

$$\begin{aligned}\frac{\partial l}{\partial \alpha_j} &= \frac{1}{2}\sum_{i=1}^{n} \boldsymbol{x}_i^T K_i^{-1}\frac{\partial K_i}{\partial \alpha_j}K_i^{-1}\boldsymbol{x}_i - \mathrm{tr}(K_i^{-1}\frac{\partial K_i}{\partial \alpha_j}) \\ &= \frac{1}{2}\sum_{i=1}^{n} \mathrm{tr}\left((\gamma_i\gamma_i^T - K_i^{-1})\frac{\partial K_i}{\partial \alpha_j}\right)\end{aligned} \tag{C.4}$$

Where $\gamma_i = K_i^{-1}\boldsymbol{x}_i$. Once $\hat{\boldsymbol{\alpha}}$ (and $\hat{K}$) is obtained, the posterior distribution for $\boldsymbol{\theta}_i$ is given by

$$\boldsymbol{\theta}_i|\boldsymbol{x}_i, \hat{K} \sim \mathcal{N}_{d_i}\left((S_i^{-1} + \hat{K}^{-1})^{-1}S_i^{-1}\boldsymbol{x}_i, (S_i^{-1} + \hat{K}^{-1})^{-1}\right) \tag{C.5}$$

Appendix D. ELBO Calculation

In the CAVI algorithm described in section 2.2.2, each update step to the parameters does not explicitly compute the objective function itself. Being able to calculate the objective function F_{ELBO} could be beneficial for monitoring the convergence of the optimization algorithm, especially since F_{ELBO} can be highly nonconcave. Here, we provide the procedure for computing F_{ELBO} using the same decomposition as in equation (B.9). The first term $\sum_{t=t_1}^{t_D} E[\log p(Y_t|U, V_t, \boldsymbol{\eta}, \mathbb{K})]$ is straightforward to compute, given the factor distributions $q(U, \mathbb{V})$. The other two terms can be identified as $\sum_{r=1}^{R} -KL(q(U_r)\|g(U_r))$ and $\sum_{r=1}^{R} -KL(q(\mathbb{V}_r)\|g(\mathbb{V}_r))$, each r'th term from either sums can be found as follows from the EBRPM subproblem:

$$\begin{aligned}\log p(\boldsymbol{x}|\hat{K}) = F_0(\hat{K}, \hat{q}(\boldsymbol{\theta})) &= E_q[\log \frac{p(\boldsymbol{x}, \boldsymbol{\theta}|\hat{K}, \boldsymbol{S})}{\hat{q}(\boldsymbol{\theta})}] \\ &= E_q[\log p(\boldsymbol{x}|\boldsymbol{\theta})] + E_q[\log \frac{\hat{g}(\boldsymbol{\theta})}{\hat{q}(\boldsymbol{\theta})}] \\ E_q[\log \frac{\hat{g}(\boldsymbol{\theta})}{\hat{q}(\boldsymbol{\theta})}] &= l(\hat{g}) + 1/2\sum_{i=1}^{n}\Big\{d_i\log(2\pi) + \log|S_i| \\ &\quad + \left(\boldsymbol{x}_i^T S_i^{-1}\boldsymbol{x}_i - 2\boldsymbol{x}_i^T S_i^{-1}E_q[\boldsymbol{\theta}_i] + \mathrm{tr}(S_i^{-1}Var_q(\boldsymbol{\theta}_i)) + E_q[\boldsymbol{\theta}_i]^T S_i^{-1}E_q[\boldsymbol{\theta}_i]\right)\Big\}\end{aligned} \tag{D.1}$$

This calculation is straightforward to implement, as the only additional quantity required is the optimized log likelihood value from the EBRPM problem, which is usually not difficult to obtain.

Appendix E. Simulations: Data Generation

In the simulation section, we set the sample size $n = 100$, variable dimension $p = 120$, and number of timepoints to be $D = 5$ complete data are generated according to the prior described in (3): For $r = 1, \ldots, R$, draw the factor matrices $\boldsymbol{u}_r \sim N(0, \eta^2 I)$, $\tilde{\boldsymbol{v}}_{ri} \sim N(0, \boldsymbol{K})$, compute the underlying structure $\boldsymbol{S}_t = \boldsymbol{U}\boldsymbol{V}_t^T$ and $\boldsymbol{X}_t = \boldsymbol{S}_t + \boldsymbol{E}_t$ where entries of $\boldsymbol{E}_t$ are standard normal draws. For the covariance structures (15), (16), we set $\sigma_v^2 = 0.2, 0.5, 1$ for the three SNR we consider and fix η at 1. In low and high correlation levels, we set $\rho = 0.2, 0.8$ for the EC structure and $\alpha = 1, 3$ for the RBF structure respectively. The free covariance structure (17) is generated as $\boldsymbol{A}^T\boldsymbol{A}/\sqrt{d}$, where $\boldsymbol{A}$ has standard normal entries. The block covariance is defined as a block diagonal covariance matrix $diag(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2)$ where $\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2$ are 2-by-2 and 3-by-3 EC covariances respectively.